The General Theory: Chapter 6

(No appendix this, that has to wait A While)

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It's late nineteen and we're talking about income, and assuredly feeling confused and bored. I'm sad to say this month's chapter probably won't help that feeling very much. The chapter itself is a jumble of terms and only we suspense is what will the last of these terms is a slightly more helpful and modern way than Keynes himself does. There's not much deep philosophical content here, just workmanlike sorting out how much income there is and how to measure it. To be fair, this was much harder in the 1930s than it is now, but that doesn't make it any more fun to read.

Of all the chapters we've talked about so far, I imagine this one reads the smoothest. There's a lot of free-floating equations, and it's hard to tell what he's driving at for most of the chapter. Once the point hits home it's then unclear – from a modern perspective at least – why the entire exercise matters. Once more, this is pretty well-understood concept nowadays, especially given how much administrative infrastructure is mentioned around, or assumed in reference to, the level of income different people have. He introduces a ton of terms that are almost never seen again in the rest of the book, and through them arrives at a national accounting identity that is taught in high schools now.

However, the tortured path it takes for a reason, because it allows Keynes to treat some properly microeconomic questions. He builds up all these different measures and terms to describe business costs to show what is in the business and what is in fact looking for other ways to make investments or employment decisions. In this, one part of this, like usual, is an argument that we are going income back to it much later on, specifically, that the "marginal efficiency of capital" is not a thing. We talked about the Appendix on User Cost and what we've worked through every one of every interest. If you'd like to chase down the issues in the interest, I recommend Perry Mehrling's great book here. If you're already, as I would expect, familiar with the definition of aggregate income, it's probably worth skipping straight to the second section.

Macroeconomics

Keynes' definition of income – all the intermediate defined terms notwithstanding – is pretty simple, the derivers it in the normal way from the producer side. Since everyone's income matches everyone's spending, it's easier accounting-wise sometimes to measure from the spending side.

As simply as possible, firms buy investment goods, intermediate goods and labor. They also sell their produced goods for more than those goods cost to make, assuming the firm is profitable. Even if some individual firm isn't profitable, at the macroeconomic level, firms in aggregate are. Since every firm selling investment or investment goods makes some profit and employs some labor, all of the money firms spend eventually turns into wages or profits somewhere down the line. The sum of all wages and profits is equal to aggregate income. Simple and familiar.

The next step is to show how this definition of income does totally fine when used as part of other national accounting identities. The value of output above the dollar value of all spending is equal to the value of income dollar value of wages and profits. Income can be spent on consumption, or saved. All output is either for consumption or for investment. Since "consumption" has the same value on both side, savings must equal investment.

That they equal one another isn't really contested by any serious economic theory – one of the main points of the *General Theory* as a whole is that investment creates an equal amount of savings. It's true that there's some goofiness about "forced savings" in Hayek and Schumpeter and Robin Barnett, but "forced savings" requires an measurement to be done in "real" divided by the price level terms. Keynes himself makes some concessions to these theories at different points, but the parts of the theory we are emphasizing here operate entirely in nominal terms.

The relationship that either of these ideas bows to the idea of a "Global Savings Glut" is really kind of a weird hack short. I'm skipping it here, because I'm in the middle of writing a piece for *Phenomenal World* about this, but the idea is basically that the level of analysis matters. There's no savings glut for the earth economy as a whole, but there are potentially savings gluts in individual economies relative to the desire for capitalists of those individual countries to pursue domestic investment. Those "savings glut" ideas find expression abroad. Whether "savings glut" is a good term for that or not I leave up to the reader, I don't much like it. Savings is, as Keynes says, at the highest level and in the final analysis, a residual.

Ultimately, Keynes arrives at the point he's discussed before, that savings is a mere residual from investment and consumption decisions. Forgets me if I noted a little sheepy describing national income identities, I've been writing a lot lately.

However, it's worth noting that method of calculating income that Keynes uses here is in some ways older than the idea of "Hang Seimens income," which I find very useful when thinking about the kinds of consumption decisions so will be talking about in later chapters.

The core idea behind Hang-Siemens income is that savings is, in the manner of a pension, defined a contribution rather than defined benefit. Rather than doing complicated adjustments to weight the impact of different mark-to-market prices on the purchasing of individual households, you just define income as "consumption + change in net wealth," and the price movements are captured in the second term. Notably, the Sonal-Pine Consumer Finance faces a number of challenges that take this basic form, because price movements in financial assets usually invoke market transactions. When one wants to think about how household consumption over time financial well-being into their consumption decisions, Hang-Simons income often provides a psychologically clearer picture than current dollar income alone.

What Keynes does in this chapter is frame current income to firms in a similar manner to Hang-Simons in order to get at the psychology of firm employment, capacity utilization, and investment decisions. Keynes's "income" is not, strange enough, because Hang-questions don't have particular relevance to the measurement of aggregate income, or the development of aggregate income as a macroeconomic concept, for fact, as I'll explain below, I'm going to hold off on a detailed explanation of the importance of "user cost" – the topic of the appendix to Chapter 6 – until after we've talked through the model of interest rates presented in Chapter 17.

While it is analytically messy to use Keynes' tortured derivation of income to justify the macroeconomic concepts of aggregate income, there is in fact a good reason why Keynes puts forward all these intermediate terms and costs in the first place. They can be quite useful as a first approximation to thinking about why firms behave in the way they do.

Depreciation: No One Knows Whats Going On

Much of the specifics of Keynes' microeconomic arguments here come from his engaging the ultimately Sisyphean task of squaring all of the relevant possible scenarios and definitions of "depreciation" into a coherent theory.

This was a huge problem for Marx, who had at least three different kinds of depreciation. The first was a kind of regular accounting depreciation, just to account for the steady loss in value of investment goods.

The next, more interesting, was the concept of "moral depreciation." In this, the value of a given capital good changes based on what other capital goods enter this produce the same commodities with different processes. If you have a machine that takes one hour an hour, and someone invents a machine that makes not chosen an hour, that certainly does something to the value of your one-hour-per-hour machine. What is done, and how it does it, and the impact of market mechanisms on the process is something of a mystery in Marx, beyond exhortations about how capitalism is great.

Probably the most famous was the version of depreciation in Marx that represented a pathway whereby the labor-that created capital goods became embodied in the commodities produced by those capital goods. Obviously some value has to transfer from the capital good to the commodity, otherwise there'd be no reason to use or capital goods when producing commodities! Plus, this has to be true for all the "production in roundabout ways of labor", stuff in Smith or Leontief, and so forth.

It's still a problem for Sraffa Pine Consistent modelling, which draws heavily from Keynes and the Post-Keynesians. Given how many different conceptual frames for depreciation there are, it's hard to make them all square with a real mathematically closed model, without having to specify everything at a per-stock-market-tick timescale.

Keynes' effort in this chapter is valiant and well-attuned to keeping the quantities of analysis both useful and in principle empirically observable. However, the task itself remains Sisyphean. The role and legal construction of straight-line accounting depreciation, accelerated depreciation, and amortization aren't really dealt with. But it's hard to blame him, those really are a permanent morass with no way out besides full repeat-expense specification and thick description of the contemporary accounting and regulatory environment.

Five Costs and a Loss

In his winding path to defining income, Keynes uses four terms to identify different intermediate costs that have some causal influence on business decisions. These are a little murky and circular in the original text, so for ease of reading we're going to break each down here.

The first is the "user cost" of a given capital good, which is equal to the carrying costs of owning that capital good less, running a room for it to sit in when it can't get rained out and the cost of the depreciation incurred by using that capital good to produce output. While this is pretty straightforward – user cost is the cost of being instead of not using – the way user cost and expectations intermingle creates a richly complex temporal system. Our last section is dedicated to exploring user cost a little more deeply.

The next is the "factor cost" of output, which is the cost paid for non-capital good inputs to production. For almost every firm, wages are part of factor cost. If you are need to buy intermediate goods as part of your production scheme, say you're a toaster company buying steel, those intermediate goods enter into factor cost as well. If user cost measures the cost of using equipment, factor cost measures the variable cost of making goods.

If you add user cost and factor cost together, you get "prime cost." The prime cost is, more or less, "how much does it cost to run and produce a given output, including expected depreciation to capital goods." This will ultimately have effects on profits, and on prices downstream, but is less central to the analysis in this chapter.

The "income of the entrepreneur" is total sales minus prime cost, or what we would call profits today. Recall, in early chapters, Keynes says that entrepreneurs make decisions about the level of employment in view of their expected income. This isn't profit maximization exactly, but rather something more like targeting profitability. What's important about the derivation here is that user cost, the factor cost are going to come back to in much greater detail later on, is embedded in profits.

The last cost category is "supplementary cost," which is essential for a particular kind of intertemporal decision-making. Firms don't, in Keynes' conceptualization, eventually-down will be large random depreciation events but basically every capital good on a long enough timeline. They also assume there will be random windfall losses – say a hurricane floods a factory, or something like that – but Keynes treats those separate from "supplementary cost."

Supplementary costs, which are expected but not really quantified, are important in the *General Theory* as a direct drag on profit. Basically if you're running a business and it breaks is bad, you have to go out and buy a new one to keep running the plant. As such, you can take that depreciation out of what you're making in the profitable run. According to Keynes, windfall loss – natural disaster or disaster in mark-to-market valuation – comes out of the firm's capital account, and Keynes's theory on how different firms react to windfall loss. It might be that the entire thing can get written off as part of a natural disaster (see 9/11. Or maybe the firm started to take a lot of depreciation first as well, and the insurance payout is more than good enough justification. All he says is that it is not necessarily the same thing as using the cost to profit on current production. All capital goods will eventually turn into dust so go absolute, but we all capital goods will eventually be hit by a crude or flood or even a hurricane.

For Keynes, the difference between supplementary cost and windfall loss is a difference in how capitalists book certain costs in their psychological ledger. When deciding how many people to hire, or how much staff to make, supplementary cost is not particularly important. Firms hire and fire based on sales minus prime cost in Keynes' model, while capitalists take supplementary cost as a bit so profits in the current period, but not in future periods. Since they expect profit to recur, changes in supplementary cost don't have a significant impact on the expectations they use to make employment decisions.

However, when capitalists are deciding how to spend their income – the profits of the economy as a whole – supplementary loss starts to matter. They will priority allocate their income in this model, and capitalists may curtail their spending until profitability returns. Through this channel, supplementary loss matters a lot more effect on hiring, but only through the action or a distance of changes to the macro environment brought about through changes in demand. If there's a crash that hits supplementary cost to employment, it's much longer and weirder than that from prime cost to employment.

Ultimately, though, it is hard to use this fine-grained specification as much more than angels on the head of a pin. Figuring out how firms actually operate in different kinds of depreciation is ultimately an empirical question, so is the consumption behavior of the people who run firms. I think it's best to read this part of the chapter as providing something like rules of thumb for how production decisions are broadly made in a monetary production economy. There's no need to go in for a priori models of agent decision-making here, and no need for a Keynesian theory to explicitly codify or quantify them.

An interesting twist to this story is the fact that Keynes has to reckon – at least briefly – with the fact that "capital goods" are a lumpy pile of physical objects, and not just a financial fund. Since there's a fair bit of product differentiation and technological change in Keynes' time, the idea that the capital stock is made up of equipment of various ages and kinds requires various techniques or different impacts to hold up. When capacity utilization rises, entrepreneurs have to make a choice which capital goods to use, and the relative depreciation costs of the different vintages of equipment are such user lumpy and unpredictable effects on user cost.

The variability of user cost also paves the way for integrating Input-Output style analysis into the Keynesian system as a way to address some conceptual quandaries with the "Schedule of the Marginal Efficiency of Capital," a concept we'll get to later concept that we have not yet run into. Without explaining the whole thing, it should suffice to say that, for the Keynesian system, the actual determination input-output coefficients matter for the "efficiency of capital."

User Cost In Particular

While the impact psychological pessimism is fun and all, the real meat of the micro side of this chapter – the concept we'll really come back to later on – is "user cost." I gave a quick definition above, but I think it's worth going into a little bit more – if metaphorical – detail.

Ivoda and noticide the Keynesian literature, it's traditional to treat the ownership of a capital asset like renting a local. It's something that has a present value determined by expected value of the discounted stream of payments it produces, just like a bond. Now, capital goods are a little bit more complicated than bonds, because you actually have to have somebody in the shop working the levers and selling the output to the new person, but the basic principle of the financial side seems the same. This is foundational to both Minsky and Tobin, for example.

However, owning capital good is only like renting a bond when the economy is operating at capacity. At capacity, everyone is running more of their capital goods, and that production is producing stable returns. Since the point of the *General Theory* is to have constant theories that apply whether or not the economy is running at capacity, it's important to examine the financial behavior of capital goods in an economy running below capacity.

Rather than a bond, renting a capital good is more like renting a call option on futures of whatever it is that that capital good produces. The option premium follows from the carrying costs of the particular capital good, and the strike price is roughly some function of (current factor cost, future factor cost, current forward price of product, expected spot price of product).

When the economy is running at capacity, this pays off like a bond. When it comes time to exercise, the future price of whatever the capital good produces is expected to be higher than the spot price of factors of production. Every option is exercised and the futures yield a dependable return. When the economy is running below capacity, however, the option representing a given capital good may not make sense of the money, and no production is undertaken. The expected future price of whatever that capital good produces is below the present factor cost. At this point, capital goods stop functioning like bonds and become much weirder. A really rigorous treatment of this is going to have to wait until after chapter 17, but we can do a quick, metaphorical treatment here.

"User cost" represents the cost of exercising the option of a capital good minus the premium cost of just selling the option. The idea is that, at the beginning of every period, the capital stock has a particular value G. In each period, capitalists can decide whether to use those capital goods or not – think of this like renting a machine or letting it sit. If they run the machine, there a certain amount of breakdown and depreciation happens. However, if they don't run the machine, there are still other maintenance and upkeep things that they have to do. You have to keep it out of the rain, and maybe run a cleaning cycle every now and again to keep the gears from seizing. The cost of running the machine might be much higher than letting it sit, but letting it sit isn't free. So the "user cost" is the cost of running the machine minus the cost of renting the machine for the same length of time without running it.

This seems obvious: having user cake means you have the option of eating it, eating your cake closes your option. You can have your cake and eat it too, but doing so means that you have to buy another cake after eating the first one. If you want to just have your cake, to make sure that you can eat it eventually, you'd have to go for a refrigerator to keep it good, and electricity for the fridge, and so on and so forth. When you have your cake, you always have the option to eat it, but that option isn't free.

The problem comes in when you lower the price of cake and the price of electricity both move around over time. If you think the cost of electricity is going to spike, then user cost of eating your cake and then buying a new one will fall dramatically. It's cheaper to eat it and then buy a new one than it is to keep the current one refrigerated long enough. If you think the cost of cake will spike, user cost flies up, assuming the cost of electricity stays the same. If it doesn't seem like being able to eat cake, and keep cakes are only going to get more expensive, you can maximize the option value your cake has by holding onto it as cake gets more and more expensive. How and when you decide what to do is always only a speculative stance, even though it involves actually owning a capital good, or a cake. Even the most "real" parts of the economy are, for Keynes, always already speculative and financial. Once we go through chapter 17, and the all the fun stuff about own-rates of own-interest, this will become much more important.

Today's post is ultimately a little more technical than it is useful, and isn't terribly exciting. Nonetheless, I think it's of a honestly just one saying "here's what this concept means, more or less. We can't use it for anything yet, sorry." In incorporation I offer a few papers on capacity utilization defaults that could be fun to think about when we start talking about how to pressure an infrastructure bill, in my opinion.

Plus, it will be fun when we actually get to use some of these ideas, someday!

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